

Lab 4 Worksheet - Practice

Q1. Extract and clean text patterns from strings using regular expressions.

You are given a messy dataset of energy sources with inconsistent units and spacing.

```
energy_sources <- c("Solar (MWh)", "Wind (GWh)", "Coal (MWh)", "Hydro (kWh)")
```

- Use `str_detect()` to identify all energy sources that contain the substring "Wind" or "Hydro" (hint: use the `|` operator in regex).
- Extract the unit type (e.g., MWh, GWh, kWh) using `str_extract()`. Make sure your pattern handles extra spaces or hyphens.
- Clean the names to keep only the energy source name, trimming whitespace and removing parentheses.
- Create a clean tibble called `energy_clean` with two columns: `source` and `unit`.
- Discussion: Why might regex cleaning be important before performing joins or aggregations?

Q2. Joining State Datasets: Socioeconomic + Regional Data

We'll use built-in datasets to explore US state-level data. `state.x77` contains metrics such as population, income, and life expectancy. `state.region` classifies each state into one of four regions: Northeast, South, North Central, West.

```
# Load datasets  
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

```
The following objects are masked from 'package:base':
```

```
intersect, setdiff, setequal, union
```

```

# Convert matrix to data frame
states <- data.frame(state.x77)

# Add state names as a new column
states$state <- rownames(states)

# select and reorder columns
states <- states |>
  select(state, Income, Population, Illiteracy)
# table for regions
regions <- data.frame(state = state.name, region = state.region)

```

- Perform a left join to combine socioeconomic and regional data.
- Perform an inner join. How many rows remain, and why is it the same or different?
- Create a summary table that shows average Income and average Life Exp by region.
- Perform an anti join to check if any state names don't match between datasets. What might cause mismatches like this in real projects?
- Discussion: Why would you prefer a left join over an inner join when merging state-level datasets?
- Use `pivot_longer()` to make the socioeconomic data longer — convert the numeric columns into key-value pairs.

Q3. Mapping

Let's visualize data by state.

```

library(ggplot2)
library(maps)
library(dplyr)

us_map <- map_data("state")

state_data <- states |>
  left_join(regions, by = "state") |>
  mutate(state = tolower(state))

```

Use a join to combine the `us_map` with the `state_data`

```
us_map_joined <- ...
```

Create a chloropleth map:

```
ggplot(us_map_joined, aes(long, lat, group = group, fill = Income)) +  
  geom_polygon(color = "white") +  
  coord_fixed(1.3) +  
  theme_minimal() +  
  scale_fill_viridis_c() +  
  labs(title = "Average Income by State")
```

b. Swap Income for Life Exp to visualize life expectancy instead. Try changing the color scale using `scale_fill_gradient()` or `scale_fill_gradient2()`

Q4. Quarto Dashboard Setup

Create a new file `ev-dashboard.qmd` and add:

```
---  
title: "EV Power Dashboard"  
format: dashboard  
---  
  
Add some sections:  
  
### Overview  
  
Your project introduction.  
  
### Map Visualization  
  
# Paste your ggplot map code here  
  
### Data Table  
  
# Show your joined dataset  
  
head(us_map_joined)  
  
Then render:  
  
quarto preview
```